p-Adic Transfer, Signature Tails, and Homology Manifolds: From a 1980 Binghamton Thesis to the 2026 Hilbert–Smith Arguments

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Abstract

I revisit the Hilbert–Smith p-adic exclusion problem through the lens of my 1980 Binghamton dissertation on p-adic transfer and Smith–Yang sequences, and through the two 2026 packaging arguments of OpenAI (Witt/signature lattice on S^d) and Dabrowski (integral L_4 signature tails). The classical Yang jump $\dim_{\mathbb{Z}}(M/\mathbb{Z}_p) = n+2$ is p-torsion; over $\mathbb{Q}$ the transfer is a projection and no dimension jump occurs. I give a clean diagnosis of the break in McAuley's 2001 arXiv manuscript, prove (relative to the 2026 manifold theorems) that connected polyhedral and resolvable ENR homology manifolds reduce to the topological-manifold case, and isolate the remaining crown jewel: nowhere-Euclidean Bryant–Ferry–Mio–Weinberger ENR $\mathbb{Z}$-homology manifolds with Quinn index $I(X) \neq 1$. There OpenAI's equal-parts lattice argument rephrases to kill every $I(X) \neq 0$, while Dabrowski's single-p cyclotomic detector fails whenever $p \mid I(X)$; a k-step cyclotomic tower restores p^k-divisibility. An inverse-limit trichotomy shows Raymond–Williams-style towers cannot produce a connected locally connected ENR cohomology-manifold counterexample. The unique remaining frontier is class HM-6 ($\dim_{\mathbb{Z}} X = n < \infty$, $\dim X = \infty$), where we develop a Borel-join Edwards–Walsh + Williams splicing program and verify its algebraic gates computationally and in Lean 4. Section 7 separates three layers when $X_{\mathrm{man}} = \varnothing$: chart proofs fail, signature exclusion is conditional on the ENR rephrase or k-tower, and inverse-limit *construction* is blocked—so emptiness of the manifold stratum is not a license to build an effective p-adic action. Each major statement is tagged *classical / relative / new* and carries a short reviewer checklist.

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Status of this draft. This is a working manuscript for reading, annotation, and possible co-authorship. It does **not** claim an independent complete proof of the classical Hilbert–Smith conjecture. Theorems that depend on OpenAI or Dabrowski are explicitly marked *[relative]*. Lean formalization notes appear in Appendix A.

1 Introduction: a memoir bridge

1.1 What Hilbert–Smith asks

A continuous effective action of a p-adic group $\mathbb{Z}_p$ on a connected finite-dimensional topological manifold M is conjectured to be impossible (Hilbert's fifth problem in the compact/transformation-group form; see [13, 18, 19]). Finite cyclic groups of p-power order do act freely on manifolds (lens spaces); the obstruction, if it exists, must use the infinite descending chain of open subgroups $p^j\mathbb{Z}_p \subset \mathbb{Z}_p$.

1.2 What I proved in 1980

In my dissertation [10], advised by L. F. McAuley at SUNY Binghamton, I constructed a p-adic transfer homomorphism for the inverse system of finite cyclic groups $\mathbb{Z}_{p^r} = A_p/p^r A_p$ and used it

to derive the Smith–Yang exact sequences governing $\mathbb{Z}_{p^r}$-actions on compact metric spaces. Two facts were already clear:

(a) Over a field L of characteristic $\neq p$ (in particular over $\mathbb{Q}$), the transfer $(1/p^r)\sum_{g\in\mathbb{Z}_{p^r}} g^*$ is a projection at every finite stage. Passing to the limit identifies $\check{H}^*(X/A_p;\mathbb{Q}) \xrightarrow{\sim} \check{H}^*(X;\mathbb{Q})^{A_p}$.

(b) Over $\mathbb{Z}$ and $\mathbb{F}_p$, the composite of transfer and projection is multiplication by $p^r \equiv 0 \pmod{p}$, producing the Bockstein/Smith–Gysin connecting maps that raise cohomological dimension. On a compact n-manifold (or suitable homology manifold) M, Yang's theorem and Bredon–Raymond–Williams give

$$\dim_{\mathbb{Z}}(M/A_p) = n+2, \qquad H_c^{n+2}(M/A_p;\mathbb{Z})/K \cong \mathbb{Z}[1/p]/\mathbb{Z}.$$

Raymond and Williams [16, 17] realized the same $(n+2)$-dimensional jump on an n-dimensional compactum by inverse limits of $(n+2)$-complexes with period-p^i homeomorphisms retracting onto n-skeleta. Those examples are *not* locally Euclidean and are not locally connected in the cohomology-manifold sense.

1.3 McAuley 2001 and the break

McAuley [12] attempted a full proof of Hilbert–Smith. His argument uses of M only local compactness, connectedness, lc^n, and local homology $H_n(X, X\setminus D;\mathbb{Z}_p) \cong \mathbb{Z}_p$. Because $\dim(M/A_p) = n+2$, saturated covers pulled back from the orbit space have order $> n+1$. To force unique n-cycles for a Černavskiĭ-type sum, he interposes a refinement U^m of order $n+1$. That cover **cannot** be A_p-invariant, and the resulting chain maps fail to be equivariant on the n-skeleton of the nerve. The missing $(n+1)$-chain is precisely where the Smith–Yang transfer cocycle lives. I record this as Corollary 3.3 below; it is historical diagnosis, not a replacement proof.

1.4 Two 2026 packagings

In September–October 2026 two independent arguments appeared that sit in a different family from Yang dim-raising:

- **OpenAI** [14]: after Newman chart reduction and Haar averaging into an odd sphere S^d, an integral Witt/signature lattice with fixed denominator bound L_d is divided into p^k equal parts $4/p^k$, contradicting integrality for large k.
- **Dabrowski** [11]: an integral signature *tail* in $L_4(A_G(T))$ is p-divisible by Karoubi/controlled Mayer–Vietoris, while an effective $\mathbb{Z}_p$-action realizes a class with tail 1 via a degree-one sphere control map and $\mathbb{CP}^2$ localization—contradicting $1 \in p\mathbb{Z}$.

Both share a Euclidean shell (Newman, charts, orientation, small open subgroups). Their arithmetic engines differ: equal parts in a fixed lattice versus single-p cyclotomic divisibility of a tail.

1.5 Relative-paper framing and seven contributions

What to check

ArXiv framing (read first). The exclusion results Theorems 6.2–6.3 are **conditional / relative**: assuming the 2026 topological-manifold exclusion machines of OpenAI [14] or Dabrowski [11], we determine the exact boundary of p-adic exclusion across the six classes HM-1–HM-6 of Definition 2.4. This manuscript does **not** claim a standalone classical proof of Hilbert–Smith on topological manifolds. What is new is the boundary map, the Quinn-index / k-tower arithmetic, the inverse-limit trichotomy, and the HM-6 construction program.

Under that framing the paper makes seven mathematical contributions:

C1. Quinn-index obstruction to single-p arithmetic (Theorem 6.1): on a nowhere-Euclidean BFMW ENR $\mathbb{Z}$-homology n-manifold ($n \geq 6$, $X_{\text{man}} = \varnothing$), Dabrowski's single index-p step yields only $I(X) \in p\mathbb{Z}$, which fails to contradict whenever an odd prime p divides $I(X)$ (e.g. $I(X) = 9$ for $p = 3$).

C2. k-step cyclotomic tower upgrade (Theorem 6.3 + Lean 4): replacing the single index-p subgroup by a k-step filtration forces $p^k \mid I(X)$; choosing $p^k > |I(X)|$ kills every nonzero Quinn index. Formalized in `CyclotomicTower.lean` / `QuinnIndex.lean`.

C3. OpenAI ENR rephrase for nowhere-Euclidean generalized manifolds (Theorem 6.2): bypassing source-sphere compactification by keeping $\widehat{M} = X \times S^{d-n}$ and pinching into S^d; equal-parts integrality kills every $I(X) \neq 0$.

C4. Unconditional reductions for polyhedral and nonempty-stratum homology manifolds (Theorems 5.1, 5.3): connected polyhedral L-homology manifolds have dense open $\mathrm{Homeo}(X)$-invariant manifold strata; resolvable ENRs reduce via $X \times \mathbb{R}^2$.

C5. One-page diagnosis of McAuley's 2001 break via the 1980 transfer (Corollary 3.3): an order-$(n+1)$ refinement cannot be A_p-invariant when $\dim_{\mathbb{Z}}(M/A_p) = n + 2$.

C6. Bounded-dimension inverse-limit no-go trichotomy (Theorem 8.1): free step-kernel C_p on compact bonding fibers forces Cantor fibers, an Euler/Smith contradiction, or destruction of local H_n.

C7. HM-6 infinite-covering-dimension frontier (Theorem 8.3, Proposition 8.6): the unique class escaping OpenAI ENR rephrase, Dabrowski k-tower, and bounded-dimension Euler obstruction, with an explicit Borel-join Edwards–Walsh + Williams splicing architecture verified by a symbolic harness (Gates G1–G4) and Lean 4 module `DranishnikovHM6.lean`.

2 Preliminaries

Definition 2.1 (Manifold stratum)**.** For a space X and fixed n, write X_{man} for the set of points with a neighborhood homeomorphic to $\mathbb{R}^n$. We say X is *nowhere Euclidean* if $X_{\text{man}} = \varnothing$.

Definition 2.2 (Homology / cohomology manifolds)**.** An ENR $\mathbb{Z}$-homology n-manifold is a Euclidean neighborhood retract X with $H_*(X, X \setminus \{x\}; \mathbb{Z}) \cong H_*(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ for every x. By Raymond [15], for compact ENRs this coincides with the ENR $\mathbb{Z}$-cohomology n-manifold condition. We write *generalized n-manifold* in the sense of Bryant–Ferry–Mio–Weinberger [1].

Definition 2.3 (Quinn index)**.** For a compact connected ENR $\mathbb{Z}$-homology n-manifold X ($n \geq 6$), Quinn's resolution index $I(X) \in \mathbb{Z}$ satisfies $I(X) \equiv 1 \pmod 8$. Resolvable spaces have $I(X) = 1$; BFMW construct examples with $I(X) \neq 1$ and $X_{\text{man}} = \varnothing$. The product formula $I(X \times Y) = I(X)I(Y)$ implies that if $I(X) \neq 1$, then $X \times Y$ is never a topological manifold for any generalized manifold Y with $I(Y) \equiv 1 \pmod 8$.

Definition 2.4 (Six homology-manifold classes)**. HM-1**

Non-*clc* Čech cohomology n-manifolds over L with $\operatorname{char}(L) \neq p$, without $\mathbb{F}_p$-regularity.

HM-2

Disconnected or 0-dimensional-junction $\mathbb{Z}$-homology n-manifolds (shrinking necklace).

HM-3

Connected polyhedral (simplicial/PL) L-homology n-manifolds.

HM-4

Connected resolvable ENR $\mathbb{Z}$-homology n-manifolds ($I(X) = 1$, or any connected $\mathbb{F}_p$-cm with $X_{\text{man}} \neq \varnothing$).

HM-5

Nowhere-Euclidean non-resolvable ENR $\mathbb{Z}$-homology n-manifolds ($n \geq 6$, $I(X) \neq 1$, $\dim X = n$).

HM-6

Non-ENR (Dranishnikov / Edwards–Walsh) $\mathbb{Z}$-cohomology n-manifolds with *finite cohomological dimension* $\dim_{\mathbb{Z}} X = n < \infty$ and *infinite topological covering dimension* $\dim X = \infty$ ($X_{\text{man}} = \varnothing$).

3 Classical transfer and the McAuley break

Lemma 3.1 (Transfer over $\mathbb{Q}$)**.** *[classical — Coppola 1980 / BRW] Let G be a compact p-adic group acting continuously on a compact metrizable space E. Then*

$$r^* : \check{H}^*(E/G; \mathbb{Q}) \longrightarrow \check{H}^*(E; \mathbb{Q})^G$$

is split injective. In particular there is no rational cohomological dimension jump.

Sketch. At each finite stage $\mathbb{Z}_{p^r}$, the operator $(1/p^r)\sum g^*$ is a projection onto invariants. Inverse limits preserve the splitting over $\mathbb{Q}$. □

What to check

Confirm that OpenAI's `lem:orbit-injectivity` is exactly this statement (no covering-dimension hypothesis). Confirm that Dabrowski never needs a rational dimension jump.

Lemma 3.2 (Yang jump is p-torsion)**.** *[classical — Yang / BRW / Coppola] If M is a compact connected orientable n-manifold (or suitable $\mathbb{Z}$-homology n-manifold) with a faithful continuous $\mathbb{Z}_p$-action, then $\dim_{\mathbb{Z}}(M/\mathbb{Z}_p) = n + 2$ and the extra cohomology is p-primary torsion of the form $\mathbb{Z}[1/p]/\mathbb{Z}$ up to finite kernel.*

Corollary 3.3 (McAuley break)**.** *[new as a one-page diagnosis; phenomenon classical] In the setting of [12], any cover U^m of order $n + 1$ pulled back toward M cannot be A_p-invariant when $\dim_{\mathbb{Z}}(M/A_p) = n + 2$. Consequently the chain-level maps used to produce unique n-cycles on $N(V_n^m)$ fail to be A_p-equivariant; the discrepancy is an $(n + 1)$-chain carrying the Smith–Yang transfer cocycle.*

What to check

Read McAuley Lemmas 6–8 (`arXiv:math/0103215v8`, pp. near 33 and 1460–1488 in the PDF text layer). Check that order$(U^m) = n+1$ is forced by wanting unique n-cycles, and that this contradicts A_p-invariance when the orbit space has dimension $n+2$.

4 Counterexamples on weakened hypotheses

Proposition 4.1 (Necklace counterexample)**.** *[new as an explicit HM-2 class; ingredients classical] Fix $m \geq 1$. On the i-th copy $M_i = S^{2m-1}$, let $\mathbb{Z}_p$ act via $\mathbb{Z}_p \twoheadrightarrow C_{p^i} \subset S^1$. Embed the spheres as radius 2^{-i} and shrink to a junction point x_∞. The resulting compact connected metrizable space X carries a faithful continuous $\mathbb{Z}_p$-action, but local homology at x_∞ has infinite rank, so X is not a $\mathbb{Z}$-homology manifold at the junction.*

Corollary 4.2 (Coefficient mismatch)**.** *[classical mechanism — Raymond 1961/1965] If one drops lc^n over $\mathbb{Z}$ and $\mathbb{F}_p$-cohomology-manifold regularity, Vietoris maps over $\mathbb{Q}$ may have Cantor $\mathbb{Z}_p$-fibers because $\check{H}^*(\mathbb{Z}_p;\mathbb{Q})$ is trivial in positive degrees. Neither 2026 proof leaks onto this class: both require local connectedness and Newman-type rigidity over $\mathbb{F}_p$.*

5 Polyhedral and resolvable reductions

Theorem 5.1 (Polyhedral reduction)**.** *[new packaging; classical PL local duality] Let $L \neq 0$ be a commutative ring with unit and let $X = |K|$ be a connected second-countable polyhedral L-homology n-manifold ($n \geq 1$). Then:*

(i) X_{man} contains $X \setminus |K^{(n-2)}|$ and is a connected, dense, open, Homeo(X)*-invariant topological n-manifold without boundary;*

(ii) every faithful continuous $\mathbb{Z}_p$-action on X restricts to a faithful continuous action on X_{man};

(iii) consequently both OpenAI [14] and Dabrowski [11], if accepted for topological manifolds, exclude faithful continuous $\mathbb{Z}_p$-actions on X.

Proof. In a simplicial L-homology n-manifold the link of every $(n-1)$-simplex has $\tilde{H}_0 \cong L$, hence exactly two vertices. Open stars off the $(n-2)$-skeleton are Euclidean. Connectivity of the dual $(n, n-1)$-gallery graph prevents $|K^{(n-2)}|$ from separating X. Homeomorphisms preserve X_{man}; density plus Hausdorffness makes restriction Homeo$(X) \to$ Homeo(X_{man}) injective. Apply the topological-manifold theorem on X_{man}. □

Remark 5.2 (Direct Dabrowski route on polyhedral $\mathbb{Q}$-hm)**.** Even without deleting $|K^{(n-2)}|$, Dabrowski's dual-block carrier acyclicity (his Lemma 7.4) holds verbatim on any polyhedral $\mathbb{Q}$-homology manifold: only rational link acyclicity is used.

What to check

Verify gallery connectivity for connected polyhedral homology manifolds over an arbitrary coefficient ring $L \neq 0$. Confirm that Dabrowski Lemma 7.4 really uses only $\mathbb{Q}$-link acyclicity.

Theorem 5.3 (Resolvable ENR / nonempty manifold stratum)**.** *[relative to Newman on $\mathbb{F}_p$-cm + Edwards–Quinn] Let X be a connected Hausdorff second-countable $\mathbb{F}_p$-cohomology n-manifold of finite covering dimension.*

(i) *If $X_{\mathrm{man}} \neq \varnothing$, then any faithful continuous $\mathbb{Z}_p$-action induces a faithful action of some open subgroup $p^j\mathbb{Z}_p \cong \mathbb{Z}_p$ on a Euclidean open set $O \subset X_{\mathrm{man}}$, so both 2026 arguments apply as written.*

(ii) *If $I(X) = 1$ (resolvable case), then $X \times \mathbb{R}^2$ is a topological $(n+2)$-manifold by Daverman and Edwards–Quinn, and a faithful action on X extends to $X \times \mathbb{R}^2$.*

Sketch of (i). Newman's theorem for connected $\mathbb{F}_p$-cohomology manifolds of finite covering dimension [18, 9] implies that no nontrivial p-power periodic homeomorphism fixes a nonempty open set. The OpenAI chart-reduction argument then produces a faithful small-subgroup action on a coordinate ball. □

What to check

Confirm Newman for connected $\mathbb{F}_p$-cm (Conner–Floyd Cor. 6.2 / Bredon). Confirm Daverman: $I(X) = 1 \Rightarrow X \times \mathbb{R}^2$ has DDP and is a manifold by Edwards–Quinn.

6 The crown jewel: nowhere-Euclidean ENR homology manifolds

By Theorem 5.3, the only connected $\mathbb{Z}$-homology n-manifolds not immediately reduced to a Euclidean chart are those with $X_{\mathrm{man}} = \varnothing$. Such spaces exist for every $n \geq 6$ [1], with arbitrary Quinn index $I(X) \equiv 1 \pmod 8$, $I(X) \neq 1$.

Theorem 6.1 (Dabrowski arithmetic fails when $p \mid I(X)$)**.** *[new as a delimitation of Dabrowski] Suppose one replaces the compactified coordinate sphere in Dabrowski §§8–13 by a compact connected orientable non-resolvable ENR $\mathbb{Z}$-homology n-manifold X with Quinn index $I(X)$. After Karoubi localization cutting $X \times \mathbb{CP}^2$ down to a scalar germ, the signature tail is*

$$\sigma_{\mathrm{tail}}(\alpha) = \big[(I(X), I(X), I(X), \ldots)\big].$$

Dabrowski's divisibility uses only $\Phi_{p^{a_i}}(1) = p$, hence only single-p divisibility $\sigma_{\mathrm{tail}} \in p\mathbb{Z}^{\mathbb{N}}/\mathbb{Z}^{(\mathbb{N})}$. For every odd prime p dividing $I(X)$ (e.g. $I(X) = 9$, $p = 3$), the relation $I(X) \in p\mathbb{Z}$ is true and yields ***no contradiction****.*

Theorem 6.2 (OpenAI-style ENR rephrase)**.** *[relative to OpenAI; new as an ENR packaging] Let X be a compact connected orientable ENR $\mathbb{Z}$-homology n-manifold with a continuous $\mathbb{Z}_p$-action. Rephrase OpenAI [14] by setting $\widehat{M} = X \times S^{d-n}$ ($d > n$ odd, $d \geq 3$) as source, keeping the target test space the standard sphere S^d. Then:*

(i) *Newman holds on X as a connected $\mathbb{F}_p$-cm;*

(ii) *$\widehat{M}$ is a retract of a compact finite polyhedron, so derived pushforwards land in the idempotent-complete test category $\mathcal{T}(S^d)$ without triangulating $\widehat{M}$;*

(iii) *moving the test uses Poincaré–Alexander duality on the homology manifold $W \subset \widehat{M}$ rather than sliding Euclidean balls;*

(iv) *the denominator lattice $L_d^{-1}\mathbb{Z}\overline{u}_d$ is a property of the target S^d alone, so the equal-parts identity*

$$\overline{z}_i(a) = \frac{4\,c_X}{p^k}\,\overline{u}_d$$

contradicts integrality once $p^k > 4|c_X|L_d$, for ***every*** *prime p and every $c_X = I(X) \neq 0$.*

What to check

Critical relative steps. (1) Idempotent-completeness of $\mathcal{T}(S^d)$ absorbs ENR retracts. (2) Alexander duality $\check{H}^d(W^+ \setminus q^{-1}(V); \mathbb{Q}) = 0$ replaces the sliding-ball homotopy. (3) The lattice bound L_d depends only on the target sphere. Mark WEAK if any of these is not accepted as written in OpenAI.

Theorem 6.3 (Dabrowski k-tower upgrade)**.** *[new arithmetic proposal; relative to Dabrowski geometric engine] Replace the single index-p subgroup detector in Dabrowski §8 by a k-step filtration*

$$H_j \supset H_{j+1} \supset \cdots \supset H_{j+k}$$

with k equivariant sphere detectors for the free C_p-quotients H_{j+r-1}/H_{j+r}. The k commuting nilpotent operators $\nu_r = p\,\mathrm{id} - \tau \otimes (-)$ force the integer multiplicity polynomial $F(t) \in \mathbb{Z}[t]$ to be divisible by $\prod_{r=0}^{k-1} \Phi_{p^{a_i-r}}(t)$, hence

$$p^k = \prod_{r=0}^{k-1} \Phi_{p^{a_i-r}}(1) \;\Big|\; F(1) = \sigma(V_i, b_i).$$

Choosing k with $p^k > |I(X)|$ upgrades the contradiction from $I(X) \in p\mathbb{Z}$ to $I(X) \in p^k\mathbb{Z}$, killing every Quinn index compatible with BFMW (combined with ε-Poincaré polyhedral approximations in place of a fixed triangulation).

What to check

Check that k commuting nilpotents really multiply cyclotomic factors in $\mathbb{Z}[t]$ without introducing denominators (the polynomial divisibility step is formalized and compiled in `CoppolaHS/CyclotomicTower.lean`). Check that BFMW ε-approximations supply the controlled input Dabrowski needs in place of a single triangulation. This theorem is the natural invitation point for Dabrowski as co-author or correspondent.

6.1 Plain-English guide to Theorems 6.1–6.3

Because the notation above mixes controlled L-theory, derived sheaf categories, and Quinn's surgery index, it is worth stating in plain English what Theorems 6.1, 6.2, and 6.3 actually say and why Theorems 6.2 and 6.3 are marked *[relative]*.

1. **Are Theorems 6.2 and 6.3 genuine theorems?** Yes—they are *conditional (relative) theorems*. The tag *[relative]* means that this short paper does not re-prove the full 2026 OpenAI or Dabrowski manifold machines from scratch; instead it proves that *if* the OpenAI sheaf-Witt machine on S^d holds (or *if* Dabrowski's controlled L-theory machine holds on ε-Poincaré spaces), then the exclusion of effective $\mathbb{Z}_p$-actions extends from topological manifolds to every compact connected orientable ENR $\mathbb{Z}$-homology manifold, even when $X_{\mathrm{man}} = \varnothing$.

2. **What is the "single-p gap" in Theorem 6.1?** Dabrowski's proof on a manifold extracts an integer signature σ and proves two things:

   (a) algebraically, using *one* step $H_j/H_{j+1} \cong C_p$ of the p-adic group and the cyclotomic identity $\Phi_{p^a}(1) = p$, the integer σ must be divisible by p ($\sigma \in p\mathbb{Z}$);

(b) geometrically, on a Euclidean manifold chart, $\sigma = 1$, contradicting $1 \in p\mathbb{Z}$.

On a nowhere-Euclidean BFMW homology manifold X ($X_{\text{man}} = \varnothing$), the local signature is no longer 1: it is *Quinn's resolution index* $I(X) \equiv 1 \pmod 8$ (for instance $I(X) = 9$, 25, or -7). If $I(X) = 9$ and $p = 3$, then Dabrowski's algebraic step says "9 is divisible by 3," which is *true* ($9 = 3 \times 3$) and gives **no contradiction**.

3. **What does "OpenAI ENR rephrase" (Theorem 6.2) mean?** In OpenAI's manifold proof, one picks a Euclidean coordinate chart $\mathbb{R}^n \subset M$, compactifies it into a *source* sphere S^d, and maps that source sphere to a *target* test sphere S^d through the orbit space. When $X_{\text{man}} = \varnothing$, X has no Euclidean chart, so there is no source sphere. The *ENR rephrase* is a simple geometric bypass: do *not* replace the source by a sphere. Keep $\widehat{M} = X \times S^{d-n}$ itself as the source space, and pinch a small neighborhood into the standard target sphere S^d. Because X is a compact *ENR* (a retract of a finite polyhedron B), the pushed-forward sheaf complex from $\widehat{M}$ is a direct summand of the pushforward from B, so it automatically lies in OpenAI's idempotent-complete test category $\mathcal{T}(S^d)$ on the target sphere without needing a chart or a triangulation. Instead of dividing 4 into p^k equal parts, one divides $4\,I(X)$ into p^k equal parts in a fixed denominator lattice $L_d^{-1}\mathbb{Z}$. Because k can be chosen arbitrarily large ($p^k > 4|I(X)|L_d$), this kills *every* $I(X) \neq 0$, even when $p \mid I(X)$.

4. **What does "Dabrowski k-tower upgrade" (Theorem 6.3) mean?** Instead of using a single subgroup step $H_j \supset H_{j+1}$ (which only divides by a single factor of p), use a *tower of k nested steps*

$$H_j \supset H_{j+1} \supset H_{j+2} \supset \cdots \supset H_{j+k}$$

inside $\mathbb{Z}_p$. Each of the k steps contributes a distinct cyclotomic polynomial $\Phi_{p^{a-r}}(t)$ dividing the character polynomial $F(t) \in \mathbb{Z}[t]$. Evaluating at $t = 1$ multiplies the k factors $\Phi_{p^{a-r}}(1) = p$, forcing $\sigma = I(X)$ to be divisible by p^k (compiled in Lean 4 in `CoppolaHS/CyclotomicTower.lean`). When $I(X) = 9$ and $p = 3$, taking $k = 3$ forces $27 \mid 9$, restoring the contradiction.

7 Settled exclusion vs open construction when $X_{\text{man}} = \varnothing$

The crown-jewel Theorems 6.1–6.3 and the inverse-limit Theorem 8.1 below are easy to misread as saying "every homology manifold forbids faithful $\mathbb{Z}_p$-actions" or, conversely, "when $X_{\text{man}} = \varnothing$ a counterexample is still available." Neither reading is correct. This section records the logical status explicitly.

7.1 The question

Q1. Do *all* homology / cohomology manifolds exclude effective continuous $\mathbb{Z}_p$-actions?

Q2. If not, can one still *construct* a faithful continuous $\mathbb{Z}_p$-action on a connected ENR $\mathbb{Z}$-homology manifold with $X_{\text{man}} = \varnothing$?

Proposition 7.1 (Short answers)**.** *[new packaging of prior theorems]*

(i) ***No to Q1.*** *Weakened classes admit actions: HM-1 (coefficient / lc^n mismatch) and HM-2 (shrinking necklace) are counterexample classes (Section 4). The 2026 arguments never claimed those scopes.*

(ii) ***For connected ENR $\mathbb{Z}$-homology manifolds the answer splits on X_{man} and $I(X)$:***

- *if $X_{\text{man}} \neq \varnothing$, or X is polyhedral, or $I(X) = 1$, then faithful continuous $\mathbb{Z}_p$-actions are excluded relative to the topological-manifold Hilbert–Smith theorems (Theorems 5.1, 5.3);*
- *if $X_{\text{man}} = \varnothing$ and $I(X) \neq 1$ (BFMW), then as-written OpenAI/Dabrowski chart arguments do not apply, but*
  - *accepting the OpenAI ENR rephrase (Theorem 6.2), or the Dabrowski k-tower (Theorem 6.3), again excludes every faithful continuous action;*
  - *independently, Theorem 8.1 blocks the standard constructive route (inverse limits of simplicial C_{p^i}-actions with free step kernels).*

(iii) ***What remains open*** *is only a narrow logical gap: a faithful continuous $\mathbb{Z}_p$-action on some BFMW space with $X_{\text{man}} = \varnothing$ and $I(X) \neq 1$, built by a method that is neither an inverse-limit tower of the Raymond–Williams type nor subject to the geometric inputs of Theorems 6.2 or 6.3. No such construction is known, and Theorem 8.1 explains why the usual attempt fails.*

7.2 Case $X_{\text{man}} \neq \varnothing$: exclusion is immediate

Theorem 7.2 (Chart reduction when a manifold point exists)**.** *[restatement of Theorem 5.3(i); classical Newman] Let X be a connected Hausdorff second-countable $\mathbb{F}_p$-cohomology n-manifold of finite covering dimension. If $X_{\text{man}} \neq \varnothing$, then a faithful continuous $\mathbb{Z}_p$-action on X induces a faithful continuous action of an open subgroup $p^j\mathbb{Z}_p \cong \mathbb{Z}_p$ on a Euclidean open set $O \subset X_{\text{man}}$. In particular, there is* ***no*** *remaining construction problem in this case: any purported action is already an action on a topological manifold chart, excluded by the 2026 manifold theorems (or by classical Hilbert–Smith if those are granted).*

Polyhedral spaces (Theorem 5.1) always have dense X_{man}. Resolvable ENRs with $I(X) = 1$ reduce by $X \times \mathbb{R}^2$ (Theorem 5.3(ii)) even if the Disjoint Disks Property fails on X itself.

7.3 Case $X_{\text{man}} = \varnothing$: three layers, not one

Nowhere-Euclidean BFMW spaces exist for $n \geq 6$ with $I(X) \equiv 1 \pmod 8$ and $I(X) \neq 1$ [1]. For those spaces the situation has *three* distinct layers.

Layer 1 — As-written 2026 proofs do not fire. Both OpenAI and Dabrowski begin by choosing a Euclidean chart (or a free point with a contractible Euclidean neighborhood). If $X_{\text{man}} = \varnothing$, that first step is unavailable. Theorem 6.1 records a further arithmetic failure mode for *naive* Dabrowski transplantation: single-p cyclotomic divisibility does not contradict $I(X)$ when $p \mid I(X)$. This is a delimitation, not a construction.

Layer 2 — Conditional exclusion (Theorems 6.2, 6.3).

- **OpenAI route.** Keep the *target* sphere S^d and replace the source by $\widehat{M} = X \times S^{d-n}$. Newman still holds on the connected $\mathbb{F}_p$-cm X; ENR retracts land in the idempotent-complete test category; the equal-parts lattice contradiction kills every $I(X) \neq 0$, including those divisible by p.
- **Dabrowski route.** Upgrade the single index-p detector to a k-step cyclotomic tower so that $p^k \mid \sigma$ with $p^k > |I(X)|$.

If either geometric package is accepted, then **no** faithful continuous $\mathbb{Z}_p$-action exists on any compact connected orientable ENR $\mathbb{Z}$-homology manifold—including those with $X_{\text{man}} = \varnothing$.

Layer 3 — Construction is blocked for inverse limits (Theorem 8.1). Even if one temporarily suspends Theorems 6.2 and 6.3, the standard way one might try to *build* an action—compatible simplicial C_{p^i}-actions on an inverse system E_i, as in Raymond–Williams—cannot produce a connected locally connected ENR cohomology manifold. Free step kernels on bonding fibers force Cantor failure of lc^0, Euler/Smith obstructions to acyclicity, or non-acyclic splicing that destroys local $\mathbb{Z}$-homology. The only inverse-limit escape is the HM-2 necklace, which is not a homology manifold at the junction.

Theorem 7.3 (No inverse-limit ENR counterexample when $X_{\text{man}} = \varnothing$)**.** *[corollary of Theorem 8.1; clarified for constructors] There is no inverse system of finite n-dimensional simplicial complexes with compatible C_{p^i}-actions and free step-kernel action on bonding fibers whose inverse limit is a connected locally connected ENR $\mathbb{Z}$-cohomology n-manifold (resolvable or not). In particular, one cannot construct a BFMW counterexample to p-adic exclusion by a Raymond–Williams-style tower.*

7.4 What a would-be constructor must achieve

To exhibit a faithful continuous $\mathbb{Z}_p$-action on a space X with $X_{\text{man}} = \varnothing$ that escapes this note, one would need all of the following simultaneously:

(a) X is a connected ENR $\mathbb{Z}$-homology n-manifold with $I(X) \neq 1$ (so $X \times Y$ never becomes a manifold);

(b) the action is jointly continuous and faithful;

(c) the geometric inputs of Theorem 6.2 fail (e.g. the pushed class is not equal-parts-integrable in $\mathcal{T}(S^d)$, or Newman/averaging cannot be run on X), *and* the k-tower inputs of Theorem 6.3 fail;

(d) the action is not realized by an inverse limit of the type forbidden by Theorem 8.1.

Items (a)–(b) are the BFMW wish-list; (c)–(d) are the actual burden. Until such an example appears, the correct summary is:

> *Within the finite-dimensional (ENR) class, homology manifolds with $X_{\text{man}} \neq \varnothing$, polyhedral triangulations, or $I(X) = 1$ are excluded via chart reduction, and nowhere-Euclidean BFMW spaces ($I(X) \neq 1$) are excluded relative to Theorem 6.2 (OpenAI ENR rephrase) or Theorem 6.3 (Dabrowski k-tower), while fixed-dimension simplicial inverse limits are blocked by Theorem 8.1. To search for a* ***non-necklace inverse-limit example*** *of a $\mathbb{Z}_p$-action on a homology manifold, one must drop finite* ***topological*** *dimension ($\dim X < \infty$, hence ENR) while keeping finite* ***cohomological*** *dimension ($\dim_{\mathbb{Z}} X = n < \infty$), as analyzed in Section 8 below.*

What to check

Verify that all three exclusions for $X_{\text{man}} = \varnothing$ (Theorem 6.2, Theorem 6.3, and Theorem 8.1) depend crucially on *finite topological covering dimension* ($\dim X < \infty$ / ENR / bounded-dimension complexes E_i), and that dropping $\dim X < \infty$ while preserving $\dim_{\mathbb{Z}} X = n$ is the unique way to reopen inverse-limit constructions without necklace pinch points.

7.5 OpenAI vs Dabrowski packaging (reminder)

Ingredient	OpenAI	Dabrowski
Euclidean shell	Newman + chart + averaging	Newman + free point + sphere control
Arithmetic object	Witt/signature lattice on S^d	L_4 integer signature tail
Divisibility	equal parts $4/p^k$, arbitrary k	single p via $\Phi_{p^a}(1) = p$
Realization	degree-1 pinch through orbit space	$\mathbb{CP}^2$ germ after localization
ENR $I(X) \neq 1$	rephrases (Thm 6.2)	needs k-tower (Thm 6.3)
Lean	none for this proof	large `HSFormal` tree

Shared Euclidean ingredients: Newman rigidity, local charts/orientation, arbitrarily small open subgroups of Homeo, and an integral signature lattice (Witt or L-theory). Sponges/solenoids/-Menger fail the Euclidean shell, not the arithmetic—so the obstruction, if correct, is Euclidean rather than purely dynamical. When $X_{\text{man}} = \varnothing$ the shell must be rephrased (Theorem 6.2) or arithmetically upgraded (Theorem 6.3); it is not simply abandoned.

8 Inverse-limit no-go in finite dimension and the $\dim_{\mathbb{Z}} < \infty$, $\dim = \infty$ frontier

8.1 The bounded-dimension trichotomy

Theorem 8.1 (Inverse-limit trichotomy in bounded topological dimension)**.** *[new] Suppose $X = \varprojlim(E_i, \pi_i)$ is an inverse limit of finite n-dimensional simplicial complexes with compatible simplicial actions of $C_{p^i} = \mathbb{Z}_p/p^i\mathbb{Z}_p$, as in Raymond–Williams. Let $K_i = \ker(C_{p^i} \to C_{p^{i-1}}) \cong C_p$ act on a bonding fiber $F_x = \pi_i^{-1}(x)$. At a regular free orbit of C_{p^i} on E_i, the step kernel K_i acts freely on the compact fiber F_x, forcing one of:*

Case 1: Zero-dimensional fibers *($|F_x| = p$). Then X accumulates Cantor $\mathbb{Z}_p$-fibers, fails lc^0, and is never a cohomology manifold over any ring.*

Case 2: Cell-like / $\mathbb{Q}$-acyclic connected fibers at stage i. *$\mathbb{F}_p$-acyclic free C_p-actions on finite-dimensional compacta are impossible by Smith. $\mathbb{Q}$-acyclic finite free C_p-complexes are impossible by Euler characteristic: $\chi(F_x) = p \cdot \chi(F_x/C_p) \equiv 0 \pmod p$, while acyclicity forces $\chi = 1$.*

Case 3: Connected fibers with $\chi \equiv 0 \pmod p$ and nontrivial homology in degrees $\leq n$. *Splicing non-acyclic fibers of dimension $\leq n$ (Raymond–Williams blocks, Menger μ^n) destroys local homology $H_n(X, X \setminus \{x\}) \cong \mathbb{Z}$ along the splicing locus.*

Thus no bounded-dimension tower yields a connected locally connected ENR cohomology manifold with a free step-kernel action on bonding fibers.

Corollary 8.2 (Only necklace escape in bounded topological dimension)**.** *[new] Within bounded-dimension simplicial towers, the only escape that preserves open manifold strata is to allow fixed points of K_i on bonding necks—i.e., a finite necklace/wedge of manifolds or lens spaces joining along singular points, converging to the HM-2 shrinking-necklace class of Proposition 4.1, where local homology fails at the accumulation point.*

What to check

Case 2 Euler obstruction: confirm $\chi = 1$ for $\mathbb{Q}$-acyclic finite complexes and $\chi \equiv 0 \pmod p$ under free C_p. Case 3: confirm that bounded-dimension RW/Menger splicing kills local $\mathbb{Z}$-homology.

8.2 How to escape the necklace trap: $\dim_{\mathbb{Z}} X = n < \infty$ with $\dim X = \infty$ (HM-6)

If one wants to construct a **non-necklace** inverse-limit example of a faithful $\mathbb{Z}_p$-action on a connected $\mathbb{Z}$-homology (or Čech $\mathbb{Z}$-cohomology) n-manifold $X = \varprojlim(E_i, \pi_i)$, the analysis above pinpoints the exact hypothesis that must be relaxed: *finite topological covering dimension* ($\dim X < \infty$).

Recall the fundamental dichotomy of cohomological dimension theory due to Edwards–Walsh [8], Dranishnikov [2], Mitchell [7], and Dranishnikov–Ferry–Weinberger [4] and Dydak–Walsh [3]:

- A compact metric $\mathbb{Z}$-cohomology n-manifold ($clc_{\mathbb{Z}}$, with $H^*(X, X \setminus \{x\}; \mathbb{Z}) \cong H^*(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\}; \mathbb{Z})$ at every point) is an **ENR** (equivalently, an **ANR**) *if and only if* its topological covering dimension is finite ($\dim X = n$).

- Using Dranishnikov's solution to Alexandroff's problem ($\dim_{\mathbb{Z}} X = 3$, $\dim X = \infty$) [2] and Edwards–Walsh cell-like resolutions, there exist compact connected $\mathbb{Z}$-cohomology n-manifolds (*class HM-6*) that are cell-like images of n-manifolds or generalized n-manifolds and satisfy
$$\dim_{\mathbb{Z}} X = n < \infty, \qquad \dim X = \infty, \qquad X_{\mathrm{man}} = \varnothing.$$
Such spaces are inverse limits of finite polyhedra E_i whose topological dimensions *grow without bound* ($\dim E_i = d_i \to \infty$), while the bonding maps $\pi_{i+1} : E_{i+1} \to E_i$ kill all cohomology in degrees $> n$ via Eilenberg–MacLane $K(\mathbb{Z}, n)$ / Postnikov resolution blocks.

Theorem 8.3 (Why $\dim_{\mathbb{Z}} X = n < \infty$, $\dim X = \infty$ reopens all three doors)**.** *[new] Let $X = \varprojlim(E_i, \pi_i)$ be a compact connected $\mathbb{Z}$-cohomology n-manifold with $\dim_{\mathbb{Z}} X = n < \infty$ and $\dim X = \infty$ (so X is not an ENR and $\dim E_i = d_i \to \infty$). Then:*

(i) ***Theorem 6.2 (OpenAI ENR rephrase) does not apply:*** *because $\dim X = \infty$, X cannot be embedded in any Euclidean space $\mathbb{R}^N$ and is not a retract of any finite polyhedron B. Hence $Rf_*\underline{\mathbb{C}}_{\widehat{M}}$ is not a direct summand of a polyhedral generator in $\mathcal{T}_0(S^d)$, and Verdier biduality $a_X^!\mathbb{R} \simeq \underline{\mathbb{R}}_X[n]$ loses its finite c-soft dimension hypothesis.*

(ii) ***Theorem 6.3 (Dabrowski k-tower) does not apply:*** *because $\dim X = \infty$, X is not an ANR/ENR and does not admit bounded-dimension controlled polyhedral Poincaré approximations.*

(iii) ***Case 2 of Theorem 8.1 (single-stage Euler obstruction) is bypassed:*** *when the fiber dimension $\dim F_x = d_i > n$ is allowed to grow with i, a finite complex F_x with a free C_p-action can be $\mathbb{Z}$-**acyclic through degree** $n+1$,*
$$\tilde{H}^q(F_x; \mathbb{Z}) = 0 \quad \textit{for all } 0 \le q \le n+1,$$
while satisfying $\chi(F_x) = p\,\chi(F_x/C_p) \equiv 0 \pmod p$ by placing its compensating Betti numbers purely in high degrees $q = d_i \gg n+1$.

Proof. Items (i) and (ii) follow immediately from the characterization of compact ENRs as finite-dimensional ANRs (Borsuk) and the definition of OpenAI's category $\mathcal{T}(S^d)$ as the idempotent completion of pushforwards from compact finite polyhedra. For (iii), take $F_x = S^{2m_i-1}$ with the standard free linear C_p-action and $2m_i - 1 > n + 1$: then $\tilde{H}^q(F_x;\mathbb{Z}) = 0$ for all $q \le n+1$ and $\chi(F_x) = 0 \equiv 0 \pmod p$. (Alternatively, in even top degree $2m_i > n+1$, a free C_p-complex with $\tilde{H}^q = 0$ for $q < 2m_i$ and $\tilde{H}^{2m_i}(F_x;\mathbb{Z}) \cong \mathbb{Z}^{p-1}$ has $\chi(F_x) = 1 + (p-1) = p \equiv 0 \pmod p$.) By the Leray spectral sequence of $\pi_i : E_i \to E_{i-1}$, vanishing of $\tilde{H}^q(F_x;\mathbb{Z})$ for $q \le n+1$ implies that π_i^* is an isomorphism on local cohomology in all degrees $\le n$, so splicing F_x does *not* destroy local n-manifold homology at stage i. □

8.3 Architecture and exact hurdle for a Dranishnikov–Raymond–Williams construction

Theorem 8.3 shows how to avoid both the necklace pinch points of HM-2 and the low-dimensional homology destruction of Raymond–Williams (Case 3 of Theorem 8.1): push the free-C_p fiber topology to growing dimensions $d_i \to \infty$, and kill those high-dimensional classes at later stages so that $\check{H}^q(X;\mathbb{Z}) = \varinjlim_i H^q(E_i;\mathbb{Z}) = 0$ for all $q > n$ (*pro-acyclicity*).

Proposition 8.4 (Pro-acyclicity over $\mathbb{F}_p$ vs $\mathbb{Z}$ in a growing-dimension C_{p^i}-tower)**.** *[new construction analysis] Consider an inverse system $E_0 \xleftarrow{\pi_1} E_1 \xleftarrow{\pi_2} \cdots$ with compatible C_{p^i}-actions and growing fiber dimensions $2m_i - 1 > n + 1$.*

(a) ***Mod-p pro-acyclicity is automatic ($\mathrm{cd}_p(\mathbb{Z}_p) = 1$):*** *Unlike the finite group C_p, which has $\mathrm{cd}_p(C_p) = \infty$, the profinite group $\mathbb{Z}_p = \varprojlim C_{p^i}$ has continuous mod-p cohomological dimension 1: under the quotient $q_{i+1} : C_{p^{i+1}} \twoheadrightarrow C_{p^i}$, the degree-2 Bockstein generator $u_i \in H^2(BC_{p^i};\mathbb{F}_p)$ pulls back to $q_{i+1}^*(u_i) = 0 \in H^2(BC_{p^{i+1}};\mathbb{F}_p)$, so*

$$\varinjlim_i H^q(BC_{p^i};\mathbb{F}_p) = H^q_{\mathrm{cts}}(\mathbb{Z}_p;\mathbb{F}_p) = 0 \quad \textit{for all } q \ge 2.$$

Concretely, on the free C_{p^i}-sphere $S^{2m_i-1} \subset \mathbb{C}^{m_i}$, the q_{i+1}-equivariant coordinate p-th power map

$$\phi_i(z_1,\dots,z_{m_i}) = \frac{(z_1^p,\dots,z_{m_i}^p)}{\|(z_1^p,\dots,z_{m_i}^p)\|}$$

has degree $p^{m_i} \equiv 0 \pmod p$, so $\phi_i^* = 0$ on $\tilde{H}^*(S^{2m_i-1};\mathbb{F}_p)$ in a single step!

(b) ***The integral solenoid tail and Dranishnikov killing:*** *On integral cohomology $\tilde{H}^{2m_i-1}(S^{2m_i-1};\mathbb{Z}) \cong \mathbb{Z}$, the map ϕ_i^* is multiplication by $p^{m_i} \neq 0$. Iterating ϕ_i in a fixed dimension $2m-1$ produces a p-adic solenoid with $\check{H}^{2m-1} \cong \mathbb{Z}[1/p]$ rather than $\dim_{\mathbb{Z}} X = n$. To obtain $\dim_{\mathbb{Z}} X = n$, stage $i+1$ must instead kill the integral class in $H^{2m_i-1}(E_i;\mathbb{Z})$ by routing π_{i+1} through a higher-dimensional equivariant Eilenberg–MacLane $K(\mathbb{Z}, 2m_i - 1)$ / Postnikov block of dimension $d_{i+1} > 2m_i - 1$ (the Edwards–Walsh / Dranishnikov step).*

(c) ***The Smith–Coppola constraint in the limit:*** *If the limit $X = \varprojlim E_i$ achieves $\dim_{\mathbb{Z}} X = n$, then by Universal Coefficients $\dim_{\mathbb{F}_p} X \le n < \infty$. Because P. A. Smith's fixed-point theorem holds for compact spaces of finite $\mathbb{F}_p$-cohomological dimension (even when $\dim X = \infty$), no finite subgroup C_p can act freely on an $\mathbb{F}_p$-acyclic closed subset of X. The construction relies on the fact that $\mathbb{Z}_p$ is torsion-free (so no finite C_p acts on X itself) while the orbit space $X/\mathbb{Z}_p$ absorbs my 1980 transfer / Yang torsion jump $\dim_{\mathbb{Z}}(X/\mathbb{Z}_p) = n + 2$ with top group $\mathbb{Z}[1/p]/\mathbb{Z}$.*

What to check

Workplan for a non-necklace HM-6 construction: To turn Proposition 8.4 into an explicit counterexample in class HM-6 ($\dim_{\mathbb{Z}} X = n$, $\dim X = \infty$), one must construct the equivariant Dranishnikov resolution block at step (b): given the free C_{p^i}-fiber S^{2m_i-1} at stage i, build the stage-$(i+1)$ free $C_{p^{i+1}}$-complex $E_{i+1} \xrightarrow{\pi_{i+1}} E_i$ so that $\pi_{i+1}^* : H^{2m_i-1}(E_i;\mathbb{Z}) \to H^{2m_i-1}(E_{i+1};\mathbb{Z})$ is zero while maintaining $\tilde{H}^q(F_x;\mathbb{Z}) = 0$ for all $q \le n+1$. The subsections below record the Williams obstruction, the Edwards–Walsh bypass, the Borel-join architecture, and the remaining analytic gate.

8.4 Williams's functor and the 1967 Part II torus obstruction

Williams [6] (building on Raymond–Williams [16, 17]) defines a splicing functor $K\Delta_q X$: replace every m-simplex of a barycentric subdivision K' by a building block (X_i, B_i, q_i) carrying a period-p^i map, glued along $\partial\Delta^m$. In Part II ([6], pp. 149–154), forcing the bonding map $L_{i+1} \to L_i$ to factor simplicially through the n-skeleton (Definition 3.1.1: so that $\dim X \le n$) produces a torus $T = S^1 \times \beta^+$ on which a map of type $(1,p)$ cannot respect the diagonal C_p-identification $y_2 y_3^i$.

Concretely, the diagonal circle $S' = \{(\theta, i\theta)\}$ is invariant, and a map g' forced into the 2-skeleton has bidegree $(1,p)$ on T, hence type (i,p) on S'. When $i \not\equiv 0 \pmod p$ this cannot factor through the diagonal C_p-quotient—the classical Part II no-go.

Remark 8.5 (Why Part II does not kill HM-6)**.** *[new]* The obstruction uses the n-skeleton retraction hypothesis. If bonding maps factor instead through higher cells of an Edwards–Walsh complex $EW_d(L_i, \mathbb{Z}, n)$ that kill $H^{>n}$ *cohomologically* without collapsing the $(n+1)$-skeleton geometrically, then S' can bound in the attached cells of dimension $> n$ and the bidegree contradiction disappears.

8.5 Edwards–Walsh resolutions and Dranishnikov K-theory blocks

The Edwards–Walsh complex $EW(K, \mathbb{Z}, n) \to |K|$ [8, 2, 3] replaces every simplex σ of dimension $> n$ by an Eilenberg–MacLane complex $K(\bigoplus \mathbb{Z}, n)$ whose $(n+1)$-skeleton is $\sigma^{(n)}$. This kills all cohomology in degrees $> n$ (so $\dim_{\mathbb{Z}} X \le n$) while allowing topological dimension $\dim K_i = d_i \to \infty$.

By Dranishnikov–Ferry–Weinberger [4], $\widetilde{KO}_*(K(\pi,n);\mathbb{Z}_p) = 0$ for $n \ge 3$ (and for finite π, $n \ge 2$). Bonding maps may therefore factor through finite skeletons of $EW(K_i, \mathbb{Z}, n)$ while remaining essential in mod-p K-theory, forcing every neighborhood in $X = \varprojlim K_i$ to have infinite covering dimension. Dranishnikov's polynomial criterion [5, Prop. 3.3] makes this checkable:

$$\big((y+1)^{p^{k-l}} - 1\big)^m \notin \big\langle (y+1)^{p^k} - 1,\, y^{n_i} \big\rangle \subset \mathbb{Z}[y]$$

whenever $m < p^l$ and $m\,p^{k-l} < n_i$, yielding unbounded cup-length on every open neighborhood $U \subset X$.

8.6 Borel-join equivariant Edwards–Walsh splicing

A finite group C_p cannot act freely on an infinite-dimensional $K(\mathbb{Z}, n)$ without nonvanishing Tate cohomology. The remedy is to perform Edwards–Walsh attachment via the **Borel construction** with a free odd sphere of growing dimension.

Proposition 8.6 (Borel-join HM-6 architecture)**.** *[new construction analysis] Fix $n \ge 2$ and an odd prime p. Inductively construct finite simplicial complexes E_i with free C_{p^i}-actions and bonding*

maps $\pi_{i+1} : E_{i+1} \to E_i$ *by*

$$E_{i+1} \;=\; \big(E_i \,\Delta_{q_i}\, EW_{d_i}(E_i, \mathbb{Z}, n)\big) \times_{C_{p^{i+1}}} S^{2m_{i+1}-1},$$

where $2m_{i+1} - 1 > n + 1$ *and* $d_{i+1} > d_i$*. Then at every finite stage:*

(i) ***Free action without Smith below degree*** $n+1$*:* $S^{2m_{i+1}-1}$ *is* $(2m_{i+1}-2)$*-connected with* $\chi = 0$*, so the Borel join is* $(n+1)$*-acyclic and* $\chi \equiv 0 \pmod p$ *(no Euler contradiction).*

(ii) ***Williams Part II bypass:*** *Edwards–Walsh* $(n+2)$*-cells let the diagonal circle* S' *bound without* n*-skeleton retraction (Remark 8.5).*

(iii) ***Pro-acyclicity*** $\Rightarrow \dim_{\mathbb{Z}} X = n$***:*** *coordinate* p*-powers kill mod-*p *classes in high degree; Edwards–Walsh cells kill integral classes in degrees* $> n$*, so* $\varinjlim_i H^q(E_i; \mathbb{Z}) = 0$ *for* $q > n$*.*

(iv) ***Infinite covering dimension on every neighborhood:*** *Dranishnikov's criterion supplies cup-length* $\to \infty$ *on every open star, hence* $\dim U = \infty$ *while* $\dim_{\mathbb{Z}} U = n$*.*

(v) ***BRW constraint on the orbit space:*** *any successful limit must satisfy* $\dim_{\mathbb{Z}}(X/A_p) = n+2$ *with top group injecting onto* $R_p(1) = \mathbb{Z}[1/p]/\mathbb{Z}$ *[16, 10]—a constraint, not an obstruction.*

Proof sketch. Item (i) is the standard connectivity of free linear C_p-spheres and the join formula $\operatorname{conn}(F \times_{C_p} S^{2m-1}) \geq 2m - 2$ when the action on the sphere is free. Item (ii) is Remark 8.5. Item (iii) combines Proposition 8.4(a)–(b) with the Postnikov killing cells of EW_{d_i}. Item (iv) is [5, Prop. 3.3–3.4]. Item (v) is the classical Bredon–Raymond–Williams transfer theorem, which uses only $\dim_{\mathbb{Z}} X = n < \infty$, not finite covering dimension. □

8.7 Remaining analytic gate and computational verification

Theorem 8.7 (Finite-stage HM-6 gates)**.** *[new — computational / Lean] On every finite stage of the architecture in Proposition 8.6, the following four gates hold for all tested parameters* $(n, p, \text{stages}) \in \{(5,3,4), (7,5,3), (6,3,5), (9,7,4)\}$ *(symbolic harness* `scripts/hm6_equivariant_ew_harness.py`*; loop log* `counterexamples/hilbert-smith/HM6_LOOP_LOG.md`*):*

G1
Williams torus bypass via Edwards–Walsh $(n+2)$*-cells.*

G2
Pro-acyclicity in degrees $> n$ *(integral and mod* p*) with strictly growing* $\dim E_i$*.*

G3
Local $\mathbb{Z}$*-cohomology* n*-manifold condition* $H^k \cong \delta_{k,n}\mathbb{Z}$*, with orbit-space Yang jump* $n+2$ *and* $R_p(1)$ *tower.*

G4
Dranishnikov Prop. 3.3 polynomial criterion with unbounded cup-length $\Rightarrow \dim U = \infty$*.*

The Borel-join connectivity lemma, the cup-length valuation gate $v_p(\text{coeff}_m) = m < k$*, and the HM-6 escape from* `OpenAIENRInput/DabrowskiTowerInput` *are formalized in* `lean/CoppolaHS/CoppolaHS/Dranishnik` *(Lean* `v4.35.0-rc3`*,* `lake build` *clean).*

Remark 8.8 (Single remaining analytic verification)**.** *[open geometric glue]* What remains for a complete geometric counterexample is the local $clc_{\mathbb{Z}}$ degree-n Poincaré duality *gluing* across simplex boundaries $\partial\sigma$ when the Edwards–Walsh block is pulled back over the Yang $(n{+}2)$-dimensional orbit space. Finite-stage gates G1–G4 verify the algebraic and homological scaffolding; the continuous-limit gluing is the sole outstanding geometric step.

What to check

Confirm G1–G4 on the harness for your preferred (n, p); check that `DranishnikovHM6.lean` builds; treat Remark 8.8 as the remaining research gate, not as a claimed existence theorem.

9 Summary matrix

Class	Faithful $\mathbb{Z}_p$?	Verdict
HM-1 non-*clc* / coeff. mismatch	yes (mechanism)	Outside both 2026 scopes
HM-2 necklace	yes (explicit)	Local homology fails at junction
HM-3 polyhedral	no	Thm 5.1 $\to$ manifold theorem
HM-4 resolvable / $X_{\text{man}} \neq \varnothing$	no	Thm 5.3, 7.2
HM-5 BFMW ENR, $X_{\text{man}} = \varnothing$, $I \neq 1$	no (rel. E or F)	Bounded-dim inverse limit blocked by Thm 8.1
HM-6 non-ENR, $\dim_{\mathbb{Z}} X = n$, $\dim X = \infty$	**open candidate**	Escapes E, F, G(2); Borel-join architecture Prop. 8.6; gates Thm 8.7; glue Rem. 8.8

Reading guide for HM-5 vs. HM-6. Within finite topological dimension ($\dim X = n$, i.e. ENRs, HM-5), exclusion holds relative to Theorem 6.2 or 6.3, and Theorem 8.1 blocks bounded-dimension inverse limits. Dropping finite topological dimension while keeping finite cohomological dimension ($\dim_{\mathbb{Z}} X = n$, $\dim X = \infty$, HM-6) is the unique frontier where a non-necklace inverse-limit construction is not ruled out by any of the three.

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A Lean status (honest boundary)

Dabrowski's formalization [11] was compiled locally from branch `palomar-review/module-system-20261015` at `b75523f0036bc857f31620ee18af970151036311` (vendored under `dabrowski/upstream-palomar/`), Lean `v4.35.0-rc3`, mathlib `1a547d8`, TauCeti `c7af81f`.

```
#print axioms HSFormal.hilbertSmith
'HSFormal.hilbertSmith' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

Extension modules live in `lean/CoppolaHS/`:

- `CyclotomicTower.lean` — product of k prime-power cyclotomics $\Rightarrow p^k \mid F(1)$ (Theorem 6.3 arithmetic);
- `QuinnIndex.lean` — abstract $I(X) : \mathbb{Z}$ with $I \equiv 1 \pmod 8$ and product-formula hypotheses;
- `PadicExclusionENR.lean` — conditional Theorems E/F shells;
- `ChallengeENR.lean` / `SolutionENR.lean` — statement + relative solution notes;
- `DranishnikovHM6.lean` — Borel-join connectivity, Dranishnikov cup-length valuation gate, HM-6 escape from ENR packaging (Theorem 8.7).

We do **not** claim a full controlled L-theory reimplementation for ENRs in Lean v1. Geometric BFMW inputs remain named hypotheses with paper citations. The HM-6 geometric gluing of Remark 8.8 is likewise not formalized.

B Reviewer checklist

Mark each item PASS / WEAK / FAIL. Co-author invitations may attach initials.

	Statement	Your mark
☐	Lemma 3.1 (transfer over $\mathbb{Q}$)	_____
☐	Lemma 3.2 (Yang p-torsion jump)	_____
☐	Corollary 3.3 (McAuley break)	_____
☐	Proposition 4.1 (HM-2 necklace)	_____
☐	Theorem 5.1 (polyhedral reduction)	_____
☐	Theorem 5.3 (resolvable / nonempty stratum)	_____
☐	Theorem 6.1 (single-p gap)	_____
☐	Theorem 6.2 (OpenAI ENR rephrase)	_____
☐	Theorem 6.3 (k-tower upgrade)	_____
☐	Prop. 7.1 (settled vs open)	_____
☐	Theorem 7.2 / 7.3	_____
☐	Theorem 8.1 (bounded-dim inverse-limit trichotomy)	_____
☐	Corollary 8.2 (necklace escape in bounded dim)	_____
☐	Theorem 8.3 ($\dim_{\mathbb{Z}} = n, \dim = \infty$ escape)	_____
☐	Prop. 8.4 (Dranishnikov–RW blueprint)	_____
☐	Prop. 8.6 (Borel-join EW + Williams)	_____
☐	Theorem 8.7 (G1–G4 harness / Lean)	_____
☐	Remark 8.8 (remaining $clc_{\mathbb{Z}}$ glue)	_____

Priority for first readers: Section 6.1 (plain-English guide to Theorems 6.1–6.3), Section 8.2–8.3 (the $\dim_{\mathbb{Z}} X = n$, $\dim X = \infty$ frontier for non-necklace examples), then Theorem 8.1 and Corollary 3.3.

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